Target Tracking

Algorithm Description

The tracker is initialized on the first frame using a skin color detector. The detector isolates pixels satisfying a set of RGB inequalities characteristic of human skin tone, restricts the search to the central horizontal band (20-80% of frame width) and upper vertical region (5-65% of frame height) to avoid latching onto background clutter, and computes row and column projections of the resulting skin mask. The bounding box center is taken as the centroid of the dominant projected region, and the box dimensions are derived directly from the spread of that region.

Each subsequent frame is processed by extracting a grayscale patch at every candidate location within a fixed search window (radius = 35 pixels) centered on the previous estimated position. The patch at each location is scored against the current working template using one of three matching criteria. After the best position is found, a scale refinement step tests five scale factors (0.90, 0.95, 1.0, 1.05, 1.10) at that winning position only, selecting the scale that produces the best score. This two-step approach, position first, scale second, prevents the scale search from latching onto large bright background regions.

Matching Methods

SSD (Sum of Squared Differences) computes the squared pixel-wise intensity difference between the candidate patch I and the template T:

$$D = \sum_{u,v} [I(u, v) - T(u, v)]^2$$

A lower value indicates a better match. SSD is sensitive to absolute intensity and is therefore vulnerable to any illumination change, motion blur, or appearance variation.

CC (Cross Correlation) computes the dot product between the candidate patch and the template:

$$C = \sum_{u,v} I(u, v)T(u, v)$$

A higher value indicates a better match. CC is biased toward bright regions regardless of structural content, making it unreliable as a standalone similarity measure without normalization.

NCC (Normalized Cross Correlation) mean-subtracts both patches before computing their correlation, then normalizes by their individual energies:

$$\hat{I}(u,v) = I(u,v) - \bar{I}, \hat{T}(u,v) = T(u,v) - \bar{T}$$

$$N = \sum_{u,v} \frac{\hat{I}(u,v)\hat{T}(u,v)}{\sqrt{[\sum_{u,v}\hat{I}(u,v)^2][\sum_{u,v}\hat{T}(u,v)^2]}}$$

The result is bounded in [1,1], where 1 indicates a perfect match. Mean subtraction removes sensitivity to additive illumination changes, and energy normalization removes sensitivity to contrast scaling, making NCC the most robust of the three methods.

Support Functions

- detect_face_region: skin color segmentation with spatial masking and projection-based centroid estimation for automatic initialization.
- to_gray: luminance weighted RGB to grayscale conversion.
- resize_template: bilinear resizing of the template to match a candidate scale during scale refinement.
- score_patch / is_better: unified scoring dispatch and comparison direction so the same search loop handles all three methods.
- search_best_match: two-step local exhaustive search: spatial search at fixed scale across the full window, followed by scale refinement at the best position only.
- adaptive_update: exponential moving average update of the working template: $T \leftarrow (1 - \alpha)T + \alpha P$, with $\alpha = 0.05$, applied only on non-occluded frames.
- compute_ncc: used independently during occlusion detection to compare the current patch against the locked reference template.
- make_video_pillow: assembles annotated frames into a GIF using Pillow.

Results and Discussion

All three methods initialized the bounding box at approximately the same location using the skin detector, correctly enclosing the girl's face in the first frame. The ellipse is drawn green during confident tracking and red when occlusion is detected.

NCC

NCC was the only method that tracked reliably throughout the sequence. Scores remained consistently high (0.87–0.96) across the majority of frames, indicating stable structural matches. The tracker correctly followed the girl's head through translation, rotation, and illumination change. The adaptive template update at $\alpha = 0.05$ allowed gradual appearance adaptation, for example, as the girl's head rotated without diverging from the target. The two-step scale search allowed the ellipse to grow and shrink as the girl moved closer and farther from the camera. NCC's invariance to mean intensity and contrast is the fundamental reason it outperforms the other two methods.

SSD

SSD failed to track reliably. Because it measures absolute pixel-level difference, any change in illumination, head pose, or background appearance inflates the score even when the tracker is positioned correctly over the face. This produced frequent false occlusion detections — the tracker froze in place at the last known good position rather than following the head, and once frozen, it could not recover because the template was no longer being updated. SSD is theoretically sound for tracking under controlled conditions, but breaks down quickly in a real video with any illumination variation.

CC

CC failed entirely. The printed center coordinates showed severe drift to locations far from the girl's face corners of the frame, bright wall regions, and window areas across all 500 frames. This is the expected failure mode of raw cross correlation: the score is dominated by regions of high absolute pixel intensity rather than structural similarity to the template. CC reported zero occluded frames, not because tracking was successful, but because no occlusion threshold was applied; a score with no upper bound cannot be thresholded reliably. CC's inclusion demonstrates concretely why NCC's normalization step is essential. Without it, the tracker is a brightness detector, not a face tracker.

Printed Output Interpretation

Each printed line reports the frame index, the estimated bounding box center (cx, cy) in pixels, the current ellipse dimensions reflecting any scale change, the raw matching score under the active method (SSD: lower is better; CC: higher is better; NCC: closer to 1 is better), and the tracking status. The status reflects whether the current patch passed the reference template NCC check.

Optional: Occlusion Handling

A boy enters the frame and partially occludes the girl's face approximately between frames 420 and 470. A naive approach of thresholding the working template's match score fails here because the adaptive template gradually incorporates the appearance changes caused by partial occlusion, keeping the score artificially high even as the boy's face replaces the girl's.

The approach taken is a dual template system with reference locking. Two templates are maintained simultaneously: a working template that updates adaptively at $\alpha = 0.05$ to handle natural appearance change, such as head rotation, and a reference template that is locked after 30 consecutive clean tracking frames. The reference captures a stable, unoccluded view of the girl's face early in the sequence and is never updated again.

On each frame, after the best position and scale are found, the current patch is compared against the locked reference template using NCC independently of the working template score. If this reference NCC falls below 0.60, the frame is flagged as occluded. On flagged frames, both the position and the working template update are frozen, preventing the occluder from corrupting the stored appearance model. Once the boy passes and the girl's face reappears, the reference NCC recovers above the threshold, and tracking resumes from the last known good position.

This approach works because the reference encodes the specific structural appearance of the girl's face. Even if the boy also has skin and a face, his facial structure, different geometry, different hair boundary, and different texture distribution produce a meaningfully lower NCC against a template built from the girl. The method correctly identifies the occlusion window under NCC. Under SSD and CC, the reference check is less reliable for the same reasons those methods fail at tracking generally: SSD scores are dominated by illumination rather than structure, and CC has no normalization to make cross-person comparison meaningful.